Algorithm 1 FM Carrier Frequency Compliance Assessment Methodology

Require: Baseband I/Q samples $S_{BB}[n]$ spanning 88 MHz to 108 MHz
Ensure: Compliance status C for an assigned channel f_c

Define channel spacing $\Delta f_{ch} \leftarrow 200$ kHz
Define valid channels $F_{ITU} \leftarrow \{88.1, 88.3, \ldots, 107.9\}$ MHz
Define regulatory tolerance $\epsilon \leftarrow (0.5 + 2 \times 10^{-5})$ kHz
for all $f_c \in F_{ITU}$ **do**
    // Stage 1 & 2: Signal Acquisition and Channelization
    $S_{ch}[n] \leftarrow \text{ApplyBandpassFilter}(S_{BB}[n], f_c, \Delta f_{ch})$
    // Stage 3: Center Frequency Estimation (DSP)
    $W[n] \leftarrow S_{ch}[n] \times \text{BlackmanWindow}(n)$ ▷ Reduce spectral leakage
    $X[k] \leftarrow \text{FFT}(W[n], N_{FFT})$ ▷ Transform to frequency domain
    $P[k] \leftarrow |X[k]|^2$ ▷ Compute Power Spectral Density
    $k_{max} \leftarrow \arg\max_k(P[k])$ ▷ Find discrete peak bin
    $\delta_k \leftarrow \frac{1}{2}\frac{P[k_{max}-1]-P[k_{max}+1]}{P[k_{max}-1]-2P[k_{max}]+P[k_{max}+1]}$ ▷ Sub-bin parabolic interpolation
    $\hat{f}_c \leftarrow (k_{max} + \delta_k) \times \frac{F_s}{N_{FFT}} + f_{offset}$ ▷ Compute exact center frequency
    // Stage 4: Absolute Error Calculation
    $\Delta f \leftarrow |\hat{f}_c - f_c|$
    // Stage 5: Normativity Assurance Check
    if $\Delta f \leq \epsilon$ **then**
        $C \leftarrow$ True ▷ Emission is Compliant
    else
        $C \leftarrow$ False ▷ Emission is Non-Compliant
    end if
    Log evidence $(f_c, \hat{f}_c, \Delta f, C)$ for reporting
end for